1. One sentence verdict

The dataset is suitable for developing an initial emergency department triage prediction model, provided that key limitations including sparse complaint variables, data quality issues, potential bias, and appropriate clinical validation and governance review are addressed before modelling.

2. Dataset summary

This dataset contains de-identified emergency department (ED) patient encounters collected during the triage process. It contains approximately 225 clinical features covering patient demographics, arrival vital signs, administrative information and presenting complaints.

The target variable is the Emergency Severity Index (ESI), a five-level emergency department acuity classification:

- ESI 1: Most urgent presentations requiring immediate intervention.
- ESI 2: High-risk patients requiring rapid assessment.
- ESI 3: Moderate urgency presentations.
- ESI 4: Lower urgency presentations.
- ESI 5: Least urgent presentations.

The dataset contains several clinically relevant feature groups.

- ❖ Demographic variables include age, gender, race, ethnicity and insurance status, which describe the patient population and allow assessment of potential fairness concerns.

- ❖ Vital sign measurements include heart rate, blood pressure, respiratory rate, oxygen saturation, temperature and glucose levels, which provide information about physiological stability.

- ❖ The dataset also contains approximately 200 binary chief complaint indicators, including symptoms such as chest pain, shortness of breath and abdominal pain, representing patient presentation at triage.

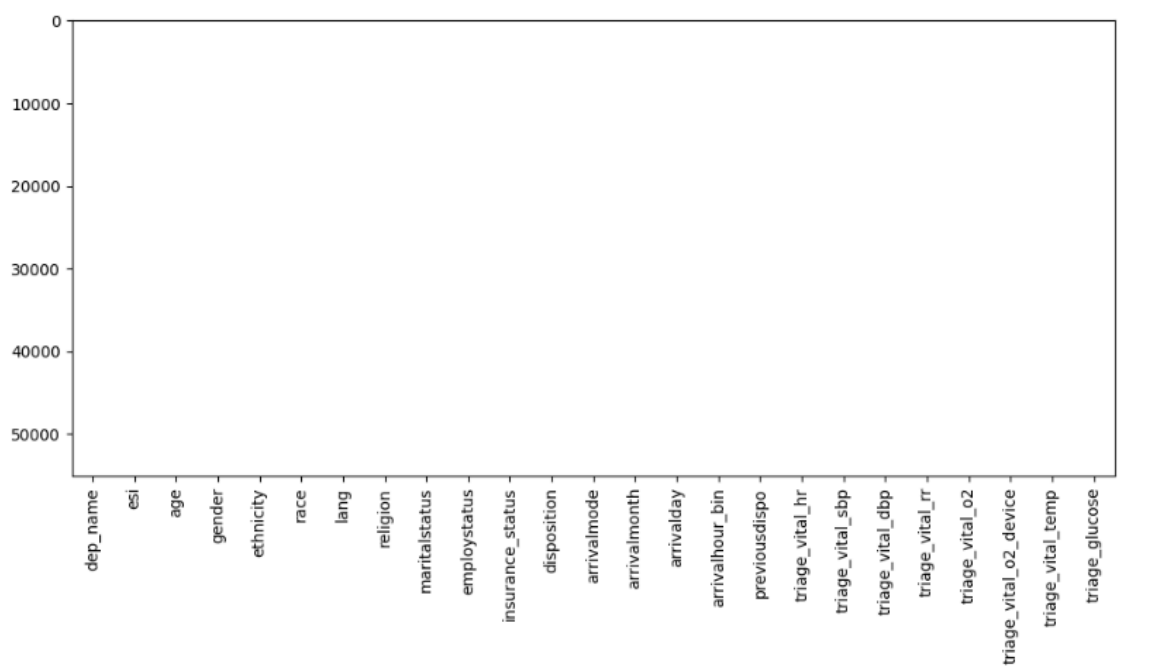
- ❖ Administrative information, including arrival mode and arrival timing, provides additional context about emergency department presentation.

The dataset reflects a real-world emergency department environment where not every measurement is necessarily available for every patient at the exact time of triage. Therefore, assessment of data completeness and quality is essential before developing a predictive model.

The exploratory analysis evaluated:

- Data completeness and missing values
- Data type consistency
- Physiological outlier detection
- Feature distributions
- Relationships between clinical features and ESI severity

Figure 1: Missingness across structured clinical variables before data cleaning.



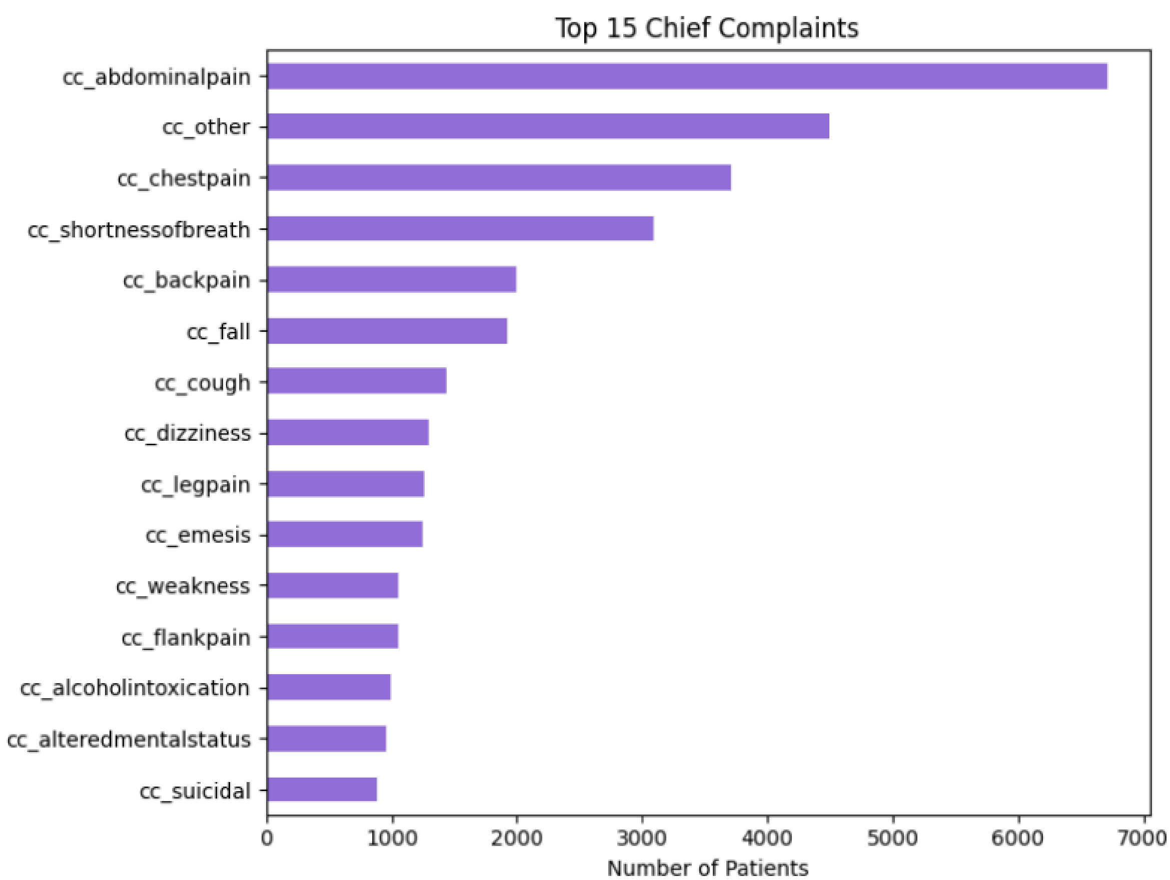
The missingness analysis demonstrated that structured clinical variables were generally well recorded within this dataset. However, data completeness should still be monitored because missingness patterns in healthcare environments may reflect clinical workflow decisions rather than random absence.

3. Top three data quality concerns

3.1. Sparse chief complaint variables

The dataset contains approximately 200 binary chief complaint features. Many of these complaints occur infrequently, creating a highly sparse feature space. Rare complaint categories may provide limited predictive information individually, while a large number of low-frequency variables may reduce model interpretability. Additionally, complaint patterns may vary between hospitals, meaning individual complaint categories may not generalise well across different healthcare settings. In order to address this issue, related complaints could be grouped into broader clinical categories. Extremely rare features could also be reviewed for removal where clinically appropriate. Model performance should be compared before and after feature aggregation to determine whether these changes improve predictive performance.

Figure 2: Distribution of the most common chief complaints recorded at triage



The figure demonstrates that a small number of complaints account for a large proportion of emergency presentations, while many complaint categories occur less frequently.

3.2. Outliers, inconsistent formatting and physiological validity

Exploratory analysis identified unusual values within several clinical measurements. These observations may represent:

- Data-entry errors.
- Incorrect recording.
- Measurement artefacts.
- Rare but genuine examples of critically ill patients.

Examples identified during quality assessment include:

- Extreme heart rate values.
- Implausible blood pressure readings.
- Temperature values requiring consideration due to unit differences.

These issues should be addressed through physiological validity checks and appropriate preprocessing. However, genuine extreme clinical values should not be removed unnecessarily, as these patients may represent the highest priority cases for a triage model.

3.3. Bias and dataset representativeness

The dataset contains demographic variables that may influence model predictions, including race, ethnicity and insurance status. The dataset also originates from a United States emergency department and may not directly represent other healthcare environments due to differences in:

- Patient populations
- Healthcare access
- Emergency department workflows
- Clinical practices

Some demographic variables may reflect social and healthcare inequalities rather than differences in clinical severity. Therefore, these variables require careful evaluation before inclusion in a predictive model.

Future work should include:

- Evaluating model performance across demographic groups
- Reviewing the role of sensitive variables
- Performing external validation using additional healthcare settings

4. Top three reasons to proceed

4.1. Strong physiological signal from triage measurements

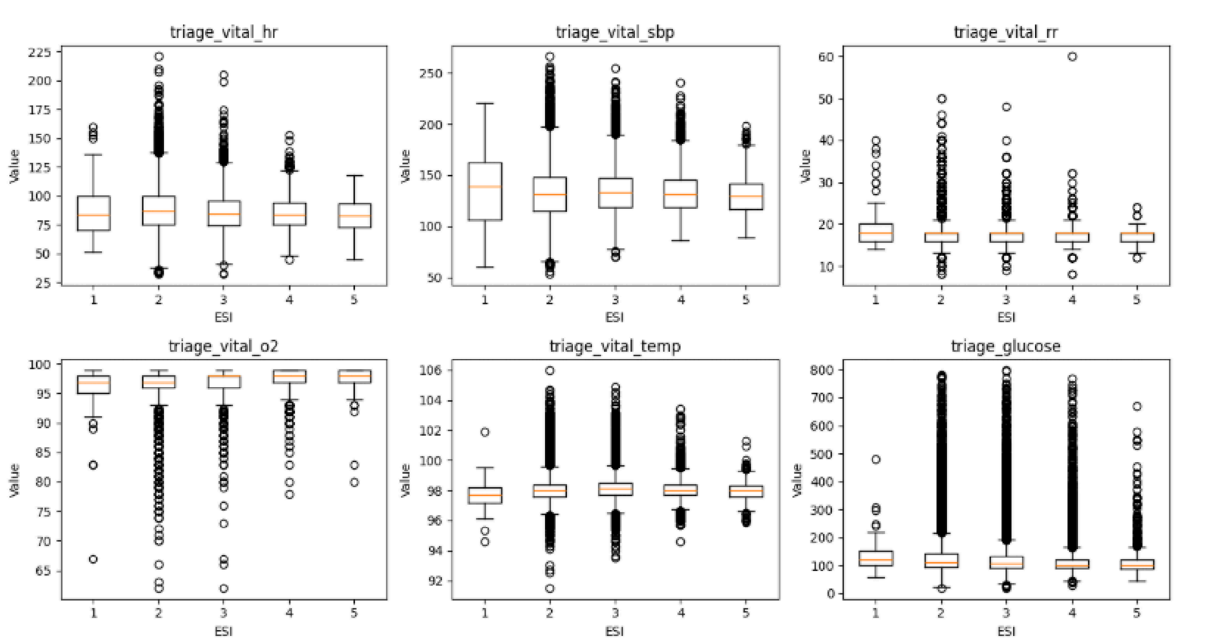
The dataset contains important physiological measurements routinely collected during emergency assessment.

Key examples include:

- Respiratory rate
- Oxygen saturation
- Heart rate
- Blood pressure
- Temperature

These measurements provide direct information about patient stability and are clinically meaningful indicators of urgency.

Figure 3: Correlation between key vital signs and Emergency Severity Index (ESI).



The correlation analysis provides an initial indication that several physiological variables demonstrate relationships with ESI severity and may contribute useful predictive information.

4.2. Broad clinical feature coverage

The dataset combines multiple sources of information available during triage, including:

- Patient characteristics
- Physiological measurements
- Presenting complaints
- Arrival information

This provides a more complete representation of patient presentation compared with relying on a single category of information. The combination of physiological and symptom-based information provides a strong foundation for exploring whether machine learning approaches can support emergency triage decisions.

4.3. Clinically meaningful prediction target

The Emergency Severity Index is an established emergency department triage framework.
The target variable provides:

- A structured severity classification
- Direct clinical interpretation
- Alignment with existing emergency department decision-making

This makes the dataset appropriate for developing an initial predictive model that can be evaluated alongside current clinical processes.

5. Key caveats

The dataset shows promise for exploratory triage modelling, but several limitations must be considered:

- The dataset originates from a US emergency department and may not generalise directly to other healthcare systems.
- Demographic variables such as race, ethnicity, and insurance status require careful evaluation because they may introduce bias.
- Only information available at the point of triage should be used. Post-triage variables such as patient disposition must be excluded to prevent data leakage.
- Temperature is recorded in Fahrenheit and should be standardised where appropriate.
- Missingness patterns may reflect emergency department workflow rather than random absence, meaning simplistic imputation approaches may introduce bias.
- The model should initially be considered a clinical decision-support prototype rather than a replacement for clinician judgement.

Further work should include:

- External validation.
- Fairness assessment.
- Clinical review.
- Monitoring after deployment.

6. Top 10 feature shortlist

The following shortlist combines exploratory correlation analysis with clinical reasoning. Features were prioritised based on their expected relationship with patient acuity and their availability during initial triage assessment.

Rank	Feature	Clinical justification
1	Respiratory Rate	Respiratory rate is one of the strongest indicators of acute physiological deterioration and demonstrated an important relationship with ESI during exploratory analysis.
2	Oxygen Saturation (SpO_2)	Oxygen saturation showed a meaningful association with ESI during exploratory analysis and is clinically recognised as an indicator of respiratory compromise.
3	Heart Rate	Abnormal heart rate may indicate shock, infection, dehydration or cardiac disease, making it highly relevant during triage.
4	Systolic Blood Pressure	Low systolic blood pressure is an important marker of circulatory compromise requiring urgent assessment.
5	Temperature	Fever or hypothermia may indicate severe infection or systemic illness requiring rapid clinical evaluation.
6	Age	Older adults generally have a greater risk of serious illness and often require higher levels of clinical attention.
7	Blood Glucose	Markedly abnormal glucose values may indicate acute diabetic emergencies or significant metabolic disturbances.
8	Chief Complaint: Chest Pain	Chest pain is frequently associated with potentially life-threatening cardiovascular conditions and typically receives higher triage priority.
9	Chief Complaint: Shortness of Breath	Respiratory complaints often indicate acute pulmonary or cardiac disease requiring prompt assessment.
10	Arrival Mode	Patients arriving by ambulance frequently represent more severe presentations and may therefore correlate with higher triage acuity.

7. Conclusion

The exploratory analysis indicates that this emergency department dataset is suitable for developing an initial triage prediction model. The dataset contains clinically meaningful variables that align with current emergency assessment practice. However, several issues must be addressed before modelling, including sparse complaint variables, physiological data quality concerns, demographic bias and validation across different healthcare settings. A future model should be considered a clinical decision-support tool designed to assist healthcare professionals rather than replace clinical judgement. Additional validation, fairness assessment and clinical review would be required before considering deployment in routine emergency care.